

Question Paper

Exam Date & Time: 21-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FIFTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

SUBJECT: ICT 3121/IT_3121 INFORMATION SECURITY

(SPL: INFORMATION TECHNOLOGY)

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) Given the 128-bit plaintext block (hex): (5)
00 11 22 33 44 55 66 77 88 99 aa bb cc dd ee ff and the AES-128 key (hex):
0f 15 71 c9 47 d9 e8 59 0c b7 ad d6 af 7f 67 98, perform the AES (first round only): (a)
AddRoundKey, (b) SubBytes, (c) ShiftRows and (d) MixColumns results (show intermediate states).

Course Outcome: "CO1" / Cognitive Level: "Application"

- 1B) A bank's online system allows customers to transfer funds, view transaction history, and download (3)
statements.
From the CIA perspective:
(a) Which component (C/I/A) would be most affected if transaction data is altered during
transmission?
(b) Which component would be most affected if the system crashes for 8 hours?
(c) Which component would be most affected if unauthorized users gain access to customer
credentials?

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 1C) Using DES S-box 1 (S1) below, find the 4-bit output for input bits 011011. (2)

DES S-box 1

	0	1	2	3	4	5	6	7	B	F
0	14	4	13	1	2	15	11	8	3	0
1	0	15	7	4	14	12	1	10	6	2
2	4	1	14	8	13	6	11	15	9	7
3	15	12	8	2	4	1	7	5	11	0

Course Outcome: "CO1" / Cognitive Level: "Application"

- 2A) In an ElGamal cryptosystem with $p=17$, $e=5$, $r=6$, $d=7$, encrypt $m=9$ and decrypt it. Explain (5)
integrity and authentication.

Course Outcome: "CO2" / Cognitive Level: "Application"

- 2B) Alice and Bob perform Diffie Hellman with $p=131$, $\alpha=7$, $a=9$, $b=12$. Compute shared secret. (3)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 2C) Compare HMAC-SHA512 and HMAC MD5. (2)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 3A) The Whirlpool algorithm produces a 512-bit message digest using the Miyaguchi-Preneel structure and an AES-based compression function, processing data in 512-bit blocks. Although it has not been extensively tested, Whirlpool is considered robust due to its AES foundation and 512-bit digest size, similar to SHA-512. However, it may be less efficient in hardware implementations. Analyze how the Miyaguchi-Preneel structure contributes to the strength of the Whirlpool algorithm and compare its design and efficiency with SHA-512. (5)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 3B) Dr. XYZ is developing a secure communication protocol for her department using the RSA digital signature scheme. (3)
She chooses $p=17$, $q=31$, and $e=94$. Generate the private key and sign the message $M=9$ and verify the signatures.

Course Outcome: "CO4" / Cognitive Level: "Application"

- 3C) For the scenario illustrated below, identify the type of forgery that has occurred in the digital signature system and justify your answer based on the attacker's capability and control over the forged message. (2)
i) After months of analysis, an attacker successfully computes Dr. Siya's **private key $a=100$** by factoring parameters from her DSS public key. The attacker can now generate valid signatures for any message exactly as Siya would.
ii) An adversary develops a mathematical algorithm that, using only Riya's **public key**, can produce a valid signature for **any message** without directly finding her private key.

Course Outcome: "CO4" / Cognitive Level: "Application"

- 4A) Compare Challenge-Response Authentication and Zero-Knowledge Proof (ZKP) protocols in terms of how they verify identity without revealing secrets. Differentiate, with a neat diagram, their verification processes by analyzing the Challenge-Response mechanism and the Guillou-Quisquater (GQ) Zero-Knowledge Protocol, highlighting their operational principles. (5)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 4B) Explain the **Feistel structure** used in DES. How does DES achieve both **confusion** and **diffusion** using the Feistel rounds? (3)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 4C) In multimedia security, (2)
a) What does compositing do?
b) What about copy/move?
Mention one way to detect each type of tampering.

Course Outcome: "CO5" / Cognitive Level: "Application"

- 5A) A user tries to access an e-commerce website using HTTPS for the first time. With a neat diagram, describe how the **TLS Handshake Protocol** operates in this scenario to **authenticate the parties** and **establish a secure session key** for encryption. (5)

Course Outcome: "CO5" / Cognitive Level: "Application"

- 5B) A large university network experiences unusual spikes in data traffic and reports of unauthorized access through Wi-Fi connections. As the network security analyst, you must choose an appropriate Intrusion Detection System (IDS) to identify and monitor these activities. Which type(s) of IDS technology would you deploy in this situation, and what factors would influence your decision? (3)

Course Outcome: "CO5" / Cognitive Level: "Application"

- 5C) In a cloud-based healthcare system, sensitive patient records are stored in encrypted form to ensure privacy. The hospital needs to **perform statistical analysis** on the encrypted data and also **search for specific patient details** without decrypting the records. **Analyze** how **Homomorphic Encryption** and **Searchable Encryption** together provide enhanced functionality and security compared to conventional encryption in this scenario. (2)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 IT_3121:INFORMATION SECURITY

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	3
4	4	CO 4	15
5	5	CO 5	12
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	12	43(86%)
4	Analysis	3	7(14%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)